

Discrete Mathematics

CS1111

Lecture 8: Number Theory

Number Theory

➤ Number theory is about **integers** ($\mathbb{Z}$) and their properties.

➤ Agenda:

- ✓ Divisibility,
- ✓ Prime numbers
- ✓ Division algorithm
- ✓ greatest common divisors,
- ✓ least common multiples
- ✓ modular arithmetic

Division

➤ If a and b are integers ($a, b \in \mathbb{Z}$) with $a \neq 0$, we say that

a **divides** b if there is an integer c so that $b = ac$.

➤ 2 divides 10 if $10 = 2 \times 5$

➤ When a divides b we say that a is a **factor** of b and that b is a **multiple** of a .

➤ The notation $a \mid b$ means that a divides b .

➤ We write $a \nmid b$ when a does not divide b

Divisibility Theorems

➤ For integers a , b , and c it is true that

if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$

Example: $3 \mid 6$ and $3 \mid 9$, so $3 \mid 15$.

if $a \mid b$, then $a \mid bc$ for all integers c

Example: $5 \mid 10$, so $5 \mid 20$, $5 \mid 30$, $5 \mid 40$, ...

if $a \mid b$ and $b \mid c$, then $a \mid c$

Example: $4 \mid 8$ and $8 \mid 24$, so $4 \mid 24$.

Primes

- A positive integer p greater than 1 is called prime if the only positive factors of p are 1 and p .
- A positive integer that is greater than 1 and is not prime is called composite.
- The fundamental theorem of arithmetic:
 - Every positive integer can be written **uniquely** as the **product of primes**, where the prime factors are written in order of increasing size.

Primes

► Examples:

$$15 = 3 \cdot 5$$

$$48 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 3 = 2^4 \cdot 3$$

$$17 = 17$$

$$100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2^2 \cdot 5^2$$

$$512 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^9$$

$$515 = 5 \cdot 103$$

$$28 = 2 \cdot 2 \cdot 7 = 2^2 \cdot 7$$

Primes

➤ If n is a composite integer, then n has a prime divisor less than or equal $\sqrt{n}$

.

➤ This is easy to see: if n is a composite integer, it must have at least two prime divisors p_1 and p_2 such that $p_1 \times p_2 = n$.

➤ p_1 and p_2 cannot both be greater than $\sqrt{n}$, because then $p_1 \times p_2 > n$.

The Division Algorithm

➤ Let a be an integer and d a positive integer. Then there are unique integers q and r , with $0 \leq r < d$, such that $a = dq + r$.

- In the above equation,
- ✓ d is called the divisor,
 - ✓ a is called the dividend,
 - ✓ q is called the quotient, and
 - ✓ r is called the remainder.

The Division Algorithm

➤ Example:

➤ When we divide 17 by 5, we have

➤ $a = dq + r$

$$17 = 5 \cdot 3 + 2.$$

✓ 17 is the **dividend**,

✓ 5 is the **divisor**,

✓ 3 is called the **quotient**, and

✓ 2 is called the **remainder**.

The Division Algorithm

➤ **Another example:**

➤ What happens when we divide -11 by 3?

➤ Note that the remainder cannot be negative.

$$-11 = 3 \times (-4) + 1$$

✓ -11 is the dividend,

✓ 3 is the divisor,

✓ -4 is called the quotient, and

✓ 1 is called the remainder.

Greatest Common Divisors

- Let a and b be integers, not both zero.
- The largest integer d such that $d \mid a$ and $d \mid b$ is called the **greatest common divisor** of a and b .
- The greatest common divisor of a and b is denoted by $\gcd(a, b)$.
- **Example 1:** What is $\gcd(48, 72)$?
- The positive common divisors of 48 and 72 are 1, 2, 3, 4, 6, 8, 12, 16, and 24, so $\gcd(48, 72) = 24$.
- **Example 2:** What is $\gcd(19, 72)$?
- The only positive common divisor of 19 and 72 is 1, so $\gcd(19, 72) = 1$.

Greatest Common Divisors

➤ Using prime factorizations:

➤ $a = p_1^{a_1} p_2^{a_2} \dots p_n^{a_n}, \quad b = p_1^{b_1} p_2^{b_2} \dots p_n^{b_n},$

➤ where $p_1 < p_2 < \dots < p_n$ and $a_i, b_i \in \mathbf{N}$ for $1 \leq i \leq n$

➤ $\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \dots p_n^{\min(a_n, b_n)}$

➤ Example:

$$a = 60 = 2^2 3^1 5^1$$

$$b = 54 = 2^1 3^3 5^0$$

$$\gcd(a, b) = 2^1 3^1 5^0 = 6$$

Relatively Prime Integers

➤ **Definition:**

➤ Two integers a and b are **relatively prime** if $\gcd(a, b) = 1$.

➤ **Examples:**

➤ Are 15 and 28 relatively prime?

➤ Yes, $\gcd(15, 28) = 1$.

➤ Are 55 and 28 relatively prime?

➤ Yes, $\gcd(55, 28) = 1$.

➤ Are 35 and 28 relatively prime?

➤ No, $\gcd(35, 28) = 7$.

Relatively Prime Integers

➤ Definition:

➤ The integers $a_1, a_2, \dots, a_n$ are **pairwise relatively prime** if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$.

➤ Examples:

➤ Are 15, 17, and 27 pairwise relatively prime?

➤ No, because $\gcd(15, 27) = 3$.

➤ Are 15, 17, and 28 pairwise relatively prime?

➤ Yes, because $\gcd(15, 17) = 1$, $\gcd(15, 28) = 1$ and $\gcd(17, 28) = 1$.

Least Common Multiples

➤ Definition:

➤ The **least common multiple** of the positive integers a and b is the smallest positive integer that is divisible by both a and b .

➤ We denote the least common multiple of a and b by $\text{lcm}(a, b)$.

➤ Examples:

$$\text{lcm}(3, 7) = 21$$

$$\text{lcm}(4, 6) = 12$$

$$\text{lcm}(5, 10) = 10$$

Least Common Multiples

➤ Using prime factorizations:

➤ $a = p_1^{a_1} p_2^{a_2} \dots p_n^{a_n}$, $b = p_1^{b_1} p_2^{b_2} \dots p_n^{b_n}$,

➤ where $p_1 < p_2 < \dots < p_n$ and $a_i, b_i \in \mathbf{N}$ for $1 \leq i \leq n$

➤ $\text{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \dots p_n^{\max(a_n, b_n)}$

➤ Example:

$$a = 60 = 2^2 3^1 5^1$$

$$b = 54 = 2^1 3^3 5^0$$

$$\text{lcm}(a, b) = 2^2 3^3 5^1 = 4 \cdot 27 \cdot 5 = 540$$

GCD and LCM

$$a = 60 = 2^2 \cdot 3^1 \cdot 5^1$$

$$b = 54 = 2^1 \cdot 3^3 \cdot 5^0$$

$$\gcd(a, b) = 2^1 \cdot 3^1 \cdot 5^0 = 6$$

$$\text{lcm}(a, b) = 2^2 \cdot 3^3 \cdot 5^1 = 540$$

Theorem: $a \cdot b = \gcd(a, b) \cdot \text{lcm}(a, b)$

Modular Arithmetic

➤ Let a be an integer and m be a positive integer.

We denote by **$a \bmod m$** the remainder when a is divided by m .

➤ **Examples:**

$$9 \bmod 4 = 1$$

$$9 \bmod 3 = 0$$

$$9 \bmod 10 = 9$$

$$-13 \bmod 4 = 3$$

Congruences

➤ Let a and b be integers and m be a positive integer. We say that **a is congruent to b modulo m** if m divides $a - b$.

➤ We use the notation **$a \equiv b \pmod{m}$** to indicate that a is congruent to b modulo m .

➤ In other words:

$a \equiv b \pmod{m}$ if and only if $a \bmod m = b \bmod m$.

Congruences

➤ Examples:

➤ Is it true that $46 \equiv 68 \pmod{11}$?

➤ Yes, because $11 \mid (46 - 68)$.

➤ Is it true that $46 \equiv 68 \pmod{22}$?

➤ Yes, because $22 \mid (46 - 68)$.

➤ For which integers z is it true that $z \equiv 12 \pmod{10}$?

➤ It is true for any $z \in \{\dots, -28, -18, -8, 2, 12, 22, 32, \dots\}$

➤ **Theorem:** Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Congruences

➤ **Theorem:** Let m be a positive integer.
If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then
 $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

➤ **Proof:**

➤ We know that $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ implies
that there are integers s and t with
 $b = a + sm$ and $d = c + tm$.

➤ Therefore,

➤ $b + d = (a + sm) + (c + tm) = (a + c) + m(s + t)$ and

➤ $bd = (a + sm)(c + tm) = ac + m(at + cs + stm)$.

➤ Hence, $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

The Euclidean Algorithm

- The **Euclidean Algorithm** finds the **greatest common divisor** of two integers a and b .
- For example, if we want to find $\gcd(287, 91)$, we **divide** 287 by 91:
- $287 = 91 \cdot 3 + 14$
- We know that for integers a , b and c , if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
- Therefore, any divisor of 287 and 91 must also be a divisor of $287 - 91 \cdot 3 = 14$.
- Consequently, $\gcd(287, 91) = \gcd(14, 91)$.

The Euclidean Algorithm

- In the next step, we divide 91 by 14:
- $91 = 14 \cdot 6 + 7$
- This means that $\gcd(14, 91) = \gcd(14, 7)$.
- So we divide 14 by 7:
- $14 = 7 \cdot 2 + 0$
- We find that $7 \mid 14$, and thus $\gcd(14, 7) = 7$.
- **Therefore, $\gcd(287, 91) = 7$.**

The Euclidean Algorithm

➤ In **pseudocode**, the algorithm can be implemented as follows:

➤ **procedure** gcd(a, b: positive integers)

➤ $x := a$

➤ $y := b$

➤ **while** $y \neq 0$

➤ **begin**

➤ $r := x \bmod y$

➤ $x := y$

➤ $y := r$

➤ **end** {x is gcd(a, b)}

Representations of Integers

➤ Let b be a positive integer greater than 1.
Then if n is a positive integer, it can be expressed **uniquely** in the form:

$$➤ n = a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b + a_0,$$

➤ where k is a nonnegative integer,

➤ $a_0, a_1, \dots, a_k$ are nonnegative integers less than b ,

➤ and $a_k \neq 0$.

➤ **Example for $b=10$:**

$$➤ 859 = 8 \cdot 10^2 + 5 \cdot 10^1 + 9 \cdot 10^0$$

Representations of Integers

➤ **Example for b=2 (binary expansion):**

➤ $(10110)_2 = 1 \cdot 2^4 + 1 \cdot 2^2 + 1 \cdot 2^1 = (22)_{10}$

➤ **Example for b=16 (hexadecimal expansion):**

➤ (we use letters A to F to indicate numbers 10 to 15)

➤ $(3A0F)_{16} = 3 \cdot 16^3 + 10 \cdot 16^2 + 15 \cdot 16^0 = (14863)_{10}$

➤

Representations of Integers

- How can we construct the base b expansion of an integer n ?
- First, divide n by b to obtain a quotient q_0 and remainder a_0 , that is,
 - $n = bq_0 + a_0$, where $0 \leq a_0 < b$.
 - The remainder a_0 is the rightmost digit in the base b expansion of n .
- Next, divide q_0 by b to obtain:
 - $q_0 = bq_1 + a_1$, where $0 \leq a_1 < b$.
 - a_1 is the second digit from the right in the base b expansion of n . Continue this process until you obtain a quotient equal to zero.

